

AUTONOMOUSMIND V2

An AI That Takes the Job, Knows Its Limits, and Charges Itself

A cognitive task agent that accepts an owner's task, senses the world through many modalities, and routes every action through a pre-execution permission gate that can approve, veto, replace, ask the owner, wait, pause to charge, or stop.

SIM-ONLY Bounded simulation — no real actuators

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This is deliberately not an animal. There is no hunger, no foraging, no reward to chase. AutonomousMind v2 is a cognitive task agent: a human owner gives it a job, it perceives the world through a wide array of sensors, plans under task, safety, uncertainty, and energy constraints, and runs every candidate action through a pre-execution permission-and-harm gate before anything happens. It pauses to charge and resumes where it left off, learns when idle, investigates safely when there is no task, and optimises its own models while it sleeps. And the owner's task never overrides safety.

What it is

AutonomousMind v2 is the flagship cognitive model built on Jneopallium. Its safety ceiling is **SIM-ONLY**: it senses, plans, decides, and acts entirely within a deterministic simulation, with no real actuator and no external service. What distinguishes it is not survival behaviour but *governed task autonomy* — it does a defined job, manages its own energy, and can be trusted with what it must not do.

The architecture is organised as twelve cognitive systems running on three timescales, with a permission gate that every action must clear before execution. The point of the demo is to show that a genuinely autonomous field agent — one that perceives richly, plans, and acts on its own — can still be held to an auditable safety standard on every single action.

Six modes, one priority order

The agent is always in exactly one operating mode, and which mode it may enter is governed by a strict priority order. Safety sits at the top; the owner's task sits below it, never above.

► The six cognitive modes. The agent transitions between them under the priority order below.

Mode	When it applies
TASK_MODE	Executing an owner-defined task.
CHARGING_MODE	Energy is low: preserve task state, dock, monitor passively.
SLEEP_OPTIMIZATION_MODE	During charging: consolidate memory, compress models, self-test.
FREE_INVESTIGATION_MODE	No task and energy sufficient: safely explore unknown areas.
IDLE_LEARNING_MODE	No task: replay logs, refine models, calibrate sensors.
EMERGENCY_SAFE_MODE	Critical fault: stop or dock safely, preserve state, report.

PRIORITY ORDER (HIGHEST FIRST)	
1 · Hard safety	harm constraints — absolute
2 · Permission	domain, legal, and authorization constraints
3 · Owner task	the defined job — never above safety
4 · Energy	continuity and charging
5 · Free investigation	safe exploration when idle
6 · Idle learning	model improvement when idle
7 · Sleep optimization	model organisation during charging

SAFETY

The single most important rule: the owner's task can never override a safety or permission constraint. A task that demands an unsafe action does not get a faster path — it gets a veto or a clarification request.

It senses like a field robot

Perception is broad and typed. The agent ingests many sensor modalities, each carrying not just a payload but its own confidence, noise estimate, calibration status, and source health — so the world model knows how much to trust each input.

The modalities include visible camera, lidar, depth, infrared, thermal, ultraviolet, radiation, radio, radar, sound, ultrasound, magnetic field, chemical/gas, pressure, temperature, humidity, and vibration, alongside owner instructions, a clock, energy state, and self-diagnostics. Sensor fusion

merges these into a single world model and, crucially, flags contradictions between them rather than averaging them away.

Twelve cognitive systems

The systems form a pipeline from raw sensing to a logged action, with the permission gate as the mandatory checkpoint at System 9.

► The twelve cognitive systems. Perception and prediction feed planning; the permission gate sits before action.

#	System	Role
0	Sensor gateway	Receive all sensor streams, owner tasks, energy, clock, diagnostics.
1	Modality perception	Convert raw modality signals into typed features.
2	Sensor fusion & world state	Merge perception into one world model; flag contradictions.
3	Attention & task relevance	Filter by task, safety, novelty, uncertainty, owner priority.
4	Working memory / workspace	Hold the current cognitive state and active subgoal.
5	Owner task manager	Parse, decompose, track, pause/resume, ask for clarification.
6	Memory	Episodic, semantic, procedural, map, task, failure, calibration.
7	Prediction & imagination	Predict outcomes, sensor readings, energy, and risks.
8	Planning	Generate plans from task, map, sensors, energy, and safety.
9	Safety / harm / permission gate	Pre-execution checkpoint every action must clear.
10	Action selection & execution	Select the approved action; apply to simulation only.
11	Learning / investigation / sleep	Idle learning, free investigation, sleep optimization.

Three timescales

Cognition runs on three loops so that the safety-critical decisions stay cheap and immediate while slower adaptation happens off the critical path.

NOTE

Fast loop, every tick: sensing, fusion, attention, working memory, task progress, planning, the safety gate, action selection, and logging. Medium loop, every 5 ticks: prediction, task decomposition, sensor-confidence and energy planning. Slow loop, every 20 ticks: idle learning, memory consolidation, model optimization, calibration. Sleep optimization runs only during charging.

The permission gate comes before the act

System 9 is the heart of the safety design. Every candidate action the agent might take is first simulated and checked — for task permission, owner authorization, forbidden actions, physical and human safety, property and resource impact, information and privacy exposure, energy feasibility, and uncertainty — and only then is a verdict issued. Execution happens after the verdict, never before it.



Figure *The pre-execution gate. A candidate action is simulated and checked against permission and harm before any verdict that lets it run.*

What makes the gate field-ready is the range of verdicts it can return — far more than a binary allow or deny.

- **APPROVED** — the action is safe and permitted; execute it.
- **VETOED** — the action would cause harm or breach permission; block it.
- **REPLACED** — substitute a safe alternative that still serves the task.
- **ASK_OWNER** — the task is ambiguous or risky; escalate to the owner rather than guess.
- **WAIT_FOR_INFORMATION** — the situation is too uncertain to act safely; hold and gather more.
- **LOW_ENERGY_PAUSE** — energy is insufficient; pause the task and charge before failure.
- **EMERGENCY_STOP** — a critical fault demands an immediate safe stop.

ASK_OWNER, WAIT_FOR_INFORMATION, and LOW_ENERGY_PAUSE are what separate a governable field agent from a reckless one. A system that can only approve or veto must always commit; this one can defer to a human, gather more evidence, or stop to charge — and it logs the reason every time.

It manages its own energy

Energy is a first-class planning constraint, not an afterthought. When the agent predicts that a task cannot finish before energy runs low, it does not push on and fail mid-task. It preserves task state, plans a route to the charger, enters CHARGING_MODE, and — while charging — enters

SLEEP_OPTIMIZATION_MODE to consolidate memory, rebuild indexes, compress its model, prune obsolete hypotheses, and run self-tests. When charging completes, it resumes exactly the task it paused.

It uses idle time well

An agent with no current task is not an idle agent. With energy to spare and no job pending, it enters FREE_INVESTIGATION_MODE and safely explores unknown areas, collecting sensor readings and improving its world model — never performing a risky or forbidden action. With observations stored, it enters IDLE_LEARNING_MODE to replay logs, improve its prediction model, calibrate sensors, and refine task templates, emitting a model-update record so the improvement is visible.

The scenarios

Twelve deterministic scenarios exercise the agent across tasks, energy, ambiguity, safety, and privacy.

► AutonomousMind scenarios. Each is deterministic and writes a full set of traces.

Scenario	What it demonstrates
owner_task_inspection	Accepts a zone-inspection task, uses the required sensors, builds a coverage map, reports.
low_energy_task_pause_resume	Pauses the task, charges, sleep-optimises, then resumes and completes it.
free_investigation_no_task	With no task, safely explores and improves its map — no risky action.
idle_learning_from_logs	Replays stored observations and improves prediction or calibration.
sleep_optimization_during_charging	Consolidates memory, rebuilds indexes, prunes, self-tests — no external action.
sensor_conflict	Lidar and camera disagree; confidence drops; it waits for information.
radiation_anomaly	Detects a radiation anomaly, marks the hazard, avoids it, reports it.
sound_radio_investigation	Triangulates a source from passive sound and radio; reports with confidence.
unsafe_owner_task	A forbidden or risky request is vetoed or escalated — never executed.
ambiguous_task	An underspecified task triggers ASK_OWNER rather than a dangerous guess.
privacy_sensitive_region	A privacy region is blocked or redacted; only a privacy-safe summary is reported.

Scenario	What it demonstrates
emergency_safe_mode	A critical fault triggers EMERGENCY_SAFE_MODE with task state preserved.

How to run it

Each scenario runs through the real Jneopallium local path and writes a directory of line-delimited traces.

1 Run a scenario

Launch any of the twelve scenarios by name.

```
scripts/demo-autonomous-mind/run_demo.sh owner_task_inspection
```

2 Read the gate's reasoning

Inspect the safety trace to see every verdict and its reason.

```
cat target/jneopallium-autonomous-mind/unsafe_owner_task/safety_trace.jsonl
```

3 Follow the task

See task progress, pauses, and resumes.

```
cat target/jneopallium-autonomous-mind/low_energy_task_pause_resume/task_trace.jsonl
```

Every scenario produces a full record: `manifest.json`, `results.jsonl`, and separate `perception_trace`, `task_trace`, `action_trace`, `safety_trace`, `learning_trace`, `sleep_optimization_trace`, and `world_trace` streams, plus a `report.json`.

Why SIM-ONLY

An agent designed to perceive widely and act in the physical world is exactly the kind of system that must prove itself exhaustively before it touches a real actuator. The sim-only ceiling keeps every action inside a deterministic world while the permission gate, the energy management, and the mode logic are validated. The path to real robotics is to replace the simulated sensor rig and actuator with real bridges while preserving the typed signal contract, the permission gate, and the sim-only-until-certified discipline — the governance travels with the agent, not the other way around.